ASP.NET CORE RESTFUL SERVICES

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Phil.About()

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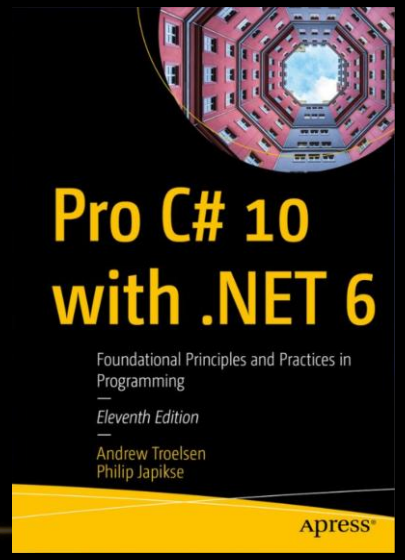

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SUPPORT LIFECYCLES

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.NET CORE SUPPORT LIFECYCLES

- Long Term Support (LTS)
  - Upgraded with patches – typically security
  - Supported for three years after GA release
  - In Support:
    - 8.0 LTS (support until 11/10/26)
    - 9.0 STS (support until 11/10/26)
    - 10.0 LTS (support until 11/14/28)
- ~~Current~~ Standard Term Support (STS)
  - Minor releases
  - Upgraded more rapidly
  - Supported for ~~6 months after the next release (LTS or STS)~~
    - 2 Years after release
  - Out of Support:
    - 5.0 support ended 05/10/22
    - 6.0 support ended 11/12/24
    - 7.0 support ended 05/14/24

https://www.microsoft.com/net/core/support

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MIGRATING FROM FRAMEWORK TO ASP.NET CORE

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ASP.NET (MVC/WEBAPI) -> ASP.NET CORE

- I Recommend:
  - File -> New project
  - Clipboard inheritance for controllers, views, and services

- Can try: https://dotnet.microsoft.com/platform/upgrade-assistant
  - Last I tried it, it didn’t handle EF Core upgrade very well

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WCF -> ASP.NET CORE

- WCF is now supported through CoreWCF
  - Open-source port of WCF (.NET F/W) to .NET

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WEB FORMS

- YARP provides an upgrade path
  - .NET application fronts the Webforms application
  - Anything not implemented in .NET is handed off to your Webforms
  - Allows for migration over time

- NOTE: I have not used this in real life

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ASP.NET CORE FUNDAMENTALS

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ASP.NET CORE

- ASP.NET Core rebuilt on top of .NET Core
- Single framework for web, services, and microservices
  - WebApi + MVC + Razor Pages + Blazor = ASP.NET Core
- Cross-platform
  - Not tied to IIS or Windows
- Takes advantage of .NET Core performance
  - Includes a high performance web server (Kestrel) built on LibUV

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ASP.NET CORE FEATURES

- Pluggable Middleware
  - Routing, authentication, static files, etc.
- Full Dependency Injection integration
- Environmental Awareness
- Simplified and Improved Configuration System
- Tag Helpers
- View Components
- Plus more!

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NOTABLE UPDATES

- ASP.NET Core 6
  - Blazor Wasm
  - CSS Isolation
  - Minimal APIs
  - “Hot” Reload
  - C# 10
- ASP.NET Core 7
  - New Middleware
  - SignalR client results
  - C# 11
- ASP.ENT Core 8
  - Blazor Web App (Server-side + Wasm)
  - Improved Authentication/Authorization
    - Blazor Identity UI
    - Identity API endpoints for JS clients
    - And more!
  - SignalR seamless reconnect
  - Keyed Services
  - C# 12

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NOTABLE UPDATES IN ASP.NET CORE 9

- Developer exception page improvements
  - Routing information, better formatting, improved readability
- Fixing 503's during app recycle
- Data protection key deletion
- Middleware support of Keyed DI
- Trust HTTPS dev cert on Linux

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NOTABLE UPDATES IN ASP.NET CORE 9 WEB APPS

- Static Asset Delivery Optimization (MVC, Razor Pages, Blazor Server)
  - Caching, Compression, + more
  - NOTE: Does not minify or bundle
- Version updates in the default template for:
  - Bootstrap (5.3.3)
  - jQuery (3.7.1)
  - jQuery Validation (1.21.0)

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NOTABLE UPDATES IN ASP.NET CORE 9 FOR SERVICES

- Minimal APIs - ProblemDetails on route groups
- OpenAPI package – Microsoft.AspNetCore.OpenApi
  - Nice progress, not quite there (no XML Comments support, no UI, versioning is still rough)
- Support for Pushed Authorization Requires (PAR)
- OIDC and Oauth parameter customization

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NOTABLE UPDATES IN ASP.NET CORE 10

- Authentication and Authorization
  - Auth metrics
  - Avoid cookie login redirects for known end points
- Performance
  - Automatic eviction from the memory pool
  - Managed memory pools
  - Plus many more…
- OpenAPI
  - OpenAPI 3.1 support
  - Response Description
  - XML doc comment support
  - Plus a ton more!
- Minimal APIs
  - Validation Support
  - Support for Server-Sent Events (SSE)
- Support for JSONPatch
- Better local URL detection
- Blazor
  - New and updated security
    - OICD, Entra ID, Windows Auth
  - Lots more

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CREATING THE WEB HOST

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ASP.NET CORE APPS ARE CONSOLE APPS

- Web host is created in top-level statements
  - Responsible for app startup and lifetime management
- At a minimum, it configures a server and request processing pipeline
- Everything is opt-in!

```
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```

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DEPENDENCY INJECTION

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ADDING CUSTOM DEPENDENCIES TO DI CONTAINER

- Configured in the Program.cs file
  - After creating the app builder and before building the app (ConfigureServices)
  - In .NET 8+, can be keyed

- Used to configure any services needed by the application
  - Transient – created each time they are requested
  - Scoped – created once per request
  - Singleton – created once

- Can plug in different IOC framework

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BUILT-IN DEPENDENCY INJECTION

- Services are accessed through:
  - Constructor injection
  - Method injection
  - [MVC/Razor] View injection
- Can also retrieve services through:
  - ApplicationServices (for non-controller classes)
  - HttpContext.RequestServices (for controllers)
  - Injection is the preferred mechanism

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INJECTING SERVICES INTO CONTROLLERS (SERVICES)

- Dependencies are added in Program.cs (ConfigureServices)

```
builder.Services.AddScoped<ICarRepo, CarRepo>();
```

- Class constructors by the DI container

```
y°εøo∫%ČcÿÿČţşóÿţøøġÿ<ICarRepo carRepø/
y°εøo∫%ČÿġcóġÕţğġø<ĬČcÿÚġyţ%∫cÿÚġyţ/
```

- ActionMethods need [FromServices] (optional for API 7+)

```
public IActionResult RazorSyntax([FromServices] ICarRepo carRepo)
```

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INJECTING KEYED SERVICES (8+)

- Dependencies are added in Program.cs (ConfigureServices)

```
builder.Services.AddKeyedScoped<ISimpleService, SimpleService>("key");
```

- Class constructors

```
y°εóo∫%ĮţšġČţşóÿţóóġÿ<.ĜÿţšMġΦġğÙġÿMo∫ġÿ</ŏġØ/#ÌÙošyóġÙġÿMo∫ġ%ÿġÿMo∫ġ/
```

- ActionMethods\PageHandlers

```
public IActionResult Index(.ĜÿţšMġΦġğÙġÿMo∫ġÿ</ŏġØ/#ÌÙošyóġÙġÿMo∫ġ%ÿġÿMo∫ġ)
```

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VALIDATE CONTROLLER SERVICES ON RUN

- Add this to Program.cs (ConfigureServices)

```
builder.Services.AddControllers().AddControllersAsServices();
```

```
builder.Host.UseDefaultServiceProvider(o =>
{
   o.ValidateOnBuild = true;
   o.ValidateScopes = true;
});
```

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ENVIRONMENTAL AWARENESS APPLICATION CONFIGURATION

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ENVIRONMENTAL AWARENESS

- ASP.NET Core uses ASPNETCORE_ENVIRONMENT variable
  - Development, Staging, and Production are built-in environments

- Environment is used throughout ASP.NET Core
  - Configuration
  - C# Execution paths
  - HTML/CSS/JS

- Simplifies deployment and testing

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APPLICATION CONFIGURATION

- Applications are configured using:
  - Default = Simple JSON
    - Other file types also supported
  - Command line arguments, Environment variables, In memory .NET objects, Encrypted user store, Custom providers

- Values are set in the order received - last in wins
  - appsettings.json
  - Environment determines which additional file to load
    - appsettings.<environment>.json

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THE OPTIONS PATTERN

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THE OPTIONS PATTERN

- Add custom classes to the services collection
- Classes are reconstituted from the application settings
- Injected using one of the IOptions<T> interfaces
  - Typically using IOptionsSnapshot or IOptionsMonitor

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USING THE OPTIONS PATTERN

- Add the option patter to the DI container

```
services.Configure<CustomSettings>(Configuration.GetSection("CustomSettings"));
```

- Inject the class into a controller constructor/action method:

```
public HomeController(IOptionsMonitor<CustomSettings> settings)
    => _settings = settings.CurrentValue;
public IActionResult Index([FromServices] IOptionsSnapshot<DealerInfo> dealerOptionsSnapshot)
    => View(dealerOptionsSnapshot.Value);
```

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JSON OPTIONS

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CONTROLLING JSON INPUT/OUTPUT

- Default casing changed
  - Classic ASP.NET Pascal cases JSON
  - ASP.NET Core camel cases JSON
  - Revert to Pascal Casing
    - JsonSerializerOptions.PropertyNamingPolicy = null;
  - Accept all casing
    - JsonSerializerOptions.PropertyNameCaseInsensitive = true;
  - JsonSerializerOptions.WriteIndented = true;
  - JsonSerializerOptions.ReferenceHandler = ReferenceHandler.IgnoreCycles;

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JSONNAMINGPOLICY OUTPUT CASING OPTIONS (8+)

- Pascal Case (PascalCase)
  - Set the naming policy to null)
- CamelCase (camelCase)
- KebabCaseLower (kebab-case-lower)
- KebabCaseUpper (Kebab-Case-Upper)
- SnakeCaseLower (snake_case_lower)
- SnakeCaseUpper (Snake_Case_Upper)

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RETURNING JSON

```
[HttpGet]
public ActionResult<IEnumerable<string>> Get()
{
  return new string[] { "value1", "value2" };
}
[HttpGet("test1")]
public IActionResult Get1()
{
  return Ok(new string[] { "value1", "value2" });
}
[HttpGet("test2")]
public string[] Get2()
{
  return new string[] { "value1", "value2" };
}
[HttpGet("test3")]
public IActionResult Get3()
{
  return new JsonResult(new string[] { "value1", "value2" });
}
```

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CONTENT NEGOTIATION

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CONTENT NEGOTIATION (BUILT-IN)

- Content will be served to the client as JSON by default
- To support XML, add:

```
builder.Services.AddControllers().AddXmlSerializerFDormatters();
```

- Send request with Accept: "application/xml“
  - Object must be serializable to XML
- Multiple supported Accept headers will be processed first-come, first-served
- To throw when the accept header is of an unsupported type:

```
builder.Services.AddControllers(options =>
{
    // Enable 406 Not Acceptable status code
    options.ReturnHttpNotAcceptable = true;
})
```

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CUSTOM CONTENT NEGOTIATION (CSV)

```
using Microsoft.AspNetCore.Mvc.Formatters;
using MediaTypeHeaderValue = Microsoft.Net.Http.Headers.MediaTypeHeaderValue;
namespace AutoLot.Api.Formatters;
public class CustomCsvOutputFormatter : TextOutputFormatter
{
    public CustomCsvOutputFormatter()
    {
        SupportedMediaTypes.Add(MediaTypeHeaderValue.Parse("text/csv"));
        SupportedEncodings.Add(Encoding.UTF8);
    }
    protected override bool CanWriteType(Type type) => typeof(IEnumerable<object>).IsAssignableFrom(type) || type.IsClass;
    public override async Task WriteResponseBodyAsync(OutputFormatterWriteContext context, Encoding selectedEncoding)
    {
        var response = context.HttpContext.Response;
        var buffer = new StringBuilder();
        var type = context.Object.GetType();
        var enumerable = context.Object as IEnumerable<object> ?? new[] { context.Object };
        var props = type.GetGenericArguments().FirstOrDefault()?.GetProperties() ?? type.GetProperties();
        // Header
        buffer.AppendLine(string.Join(",", props.Select(p => p.Name)));
        // Rows
        foreach (var item in enumerable)
        {
            var values = props.Select(p => p.GetValue(item, null)?.ToString()?.Replace(",", " ") ?? "");
            buffer.AppendLine(string.Join(",", values));
        }
        await response.WriteAsync(buffer.ToString());
    }
}
```

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DATA SHAPING

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DATA SHAPING (GET)

- Return only the selected fields listed in the request, plus the PK

```
api/DataShaping?fields=firstname,lastname,timestamp
```

- Use reflection and ExpandObjects to return the PK and the requested fields
- Reflection is expensive, and process is fragile, so consider the value

```
[HttpGet]
[Produces("application/json")]
public IActionResult GetFromQuery([FromQuery] string fields)
{
   var owners = driverRepo.GetAll(); // Assume this returns List<Owner>
   var shapedOwners = dataShaper.ShapeData(owners, fields);
   return Ok(shapedOwners);
}
```

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DATA SHAPING (POST/PUT)

- Send only specific fields in the request, plus the PK

```
/api/DataShaping/5?v=1.0&values={"FirstName":"Fred","LastName":"FlintStone"}
```

- Retrieve instance from data store
  - Option 1: Use Newtonsoft JsonConvert.PopulateObject

```
var car = _repo.Get(id);
JsonConvert.PopulateObject(values, car);
_repo.Update(car);
```

  - Option 2: Use System.Text.Json and reflection
    - NOTE: Using dynamic or expando with objects isn't recommended

  - Reflection is expensive, and the process is fragile, so consider the value

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CORS

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CROSS ORIGIN RESOURCE SHARING (CORS)

- ASP.NET Core supports CORS
  - Header
  - Method
  - Credentials
  - Origin

- Configured in the DI container with the web app builder (ConfigureServices)
- Applied to the pipeline in the web app (Configure)

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HYPERMEDIA AS THE ENGINE OF APPLICATION STATE (HATEOAS)

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HYPERMEDIA AS THE ENGINE OF APPLICATION STATE (HATEOAS)

- Provide hypermedia links in the API response
- Goal is to provide flexibility to the API
- Contrasts with “traditional” methods, like versioning and OpenAPI docs

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SIMPLE HATEOAS EXAMPLE

```
{
    "accountNumber": 10,
    "accountType": "Checking",
    "links": [
        {
            "href": "10/balance",
            "rel": "balance",
            "type" : "GET"
        },
        {
            "href": "10/withdraw",
            "rel": "withdraw",
            "type" : "POST"
        },
        {
            "href": "10/deposit",
            "rel": "deposit",
            "type" : "POST"
        }
    ]
}
```

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MODEL BINDING AND VALIDATION

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MODEL BINDING

- Route data and query string values are used only for simple types.
- Implicit model binding of complex types uses the action method parameter
  - Explicit model binding is also available

- Errors in data type conversion are added to ModelState

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MODEL BINDING

- Sources for data (in order):
  - Form fields
  - The request body (API services)
  - Route data
  - Query string parameters
  - Uploaded files

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MODEL BINDING EXPLICIT SOURCES

- Can explicitly specify the source:
  - [FromQuery] - query string.
  - [FromRoute] - route data.
  - [FromForm] - posted form fields.
  - [FromBody] - request body.
  - [FromHeader] - HTTP headers.

- Can also explicitly call for model binding with code

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VALIDATION

- Data annotations offer built-in client and server-side validation
  - Custom validation attributes are also available

- Implicit validation occurs with model binding
  - Explicit validation is also available

- Errors in validation are added to ModelState

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CUSTOM VALIDATION

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CUSTOM VALIDATION ATTRIBUTES

- Derive from ValidationAttribute for server-side validation
  - Override the IsValid method
  - ValidationContext provides access to the entire model

- Implement IClientModelValidator for client-side validation
  - AddValidation methods adds data- attributes to rendered elements
  - Ties into JS validation (e.g. Jquery validations)

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LOGGING

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LOGGING IS BUILT INTO FRAMEWORK

- LoggerFactory is part of the framework
- Extensible using 3rd party loggers (e.g. SeriLog)

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SERVICE CONTROLLERS

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CONTROLLERBASE

- Contains helper methods for:
  - Returning response messages
    - No body: NoContent (204), Ok (200), BadRequest (400)
    - With body: OkObjectResult(car) (200)
  - Model Binding and Validation
  - Redirecting requests

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ACTIONS

- Return IActionResult (Task<IActionResult>)

- GET is the default verb for actions
  - Any other verbs must be specified (browsers support GET and POST)
  - All methods should specify HTTP Verb
    - Unmarked action methods support all other methods

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APICONTROLLER ATTRIBUTE

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API CONTROLLER ATTRIBUTE

- ApiController Attribute (applied at the project or controller level)
  - Enables REST-specific behavior for controllers
    - Automatic 400 responses on model validation errors
    - Binding source parameter inference
    - Multipart/form-data inference

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BINDING SOURCE PARAMETER INFERENCE

- FromBody - complex types (one per action)
  - [FromBody(EmptyBodyBehavior = EmptyBodyBehavior.Allow)]
- FromForm - IFormFile and IFormFileCollection
- FromRoute - any values matching the route
- FromQuery - all others

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MODIFYING APICONTROLLER ATTRIBUTE FUNCTIONALITY

- Suppressions:
  - SuppressModelStateInvalidFilter = true;
    - // Automatic model state binding errors
  - SuppressInferBindingSourcesForParameters= true;
    - //All binding inference
  - SuppressConsumesConstraintForFormFileParameters = true;
    - // Multipart/form-data content type inference

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PROBLEM DETAILS

- Don't create a problem details error object if set to true
  - SuppressMapClientErrors = false;
  - https://tools.ietf.org/html/rfc7807

- Add additional information to the problem details object
  - ClientErrorMapping[StatusCodes.Status404NotFound].Link = "https://httpstatuses.com/404";
  - ClientErrorMapping[StatusCodes.Status404NotFound].Title = "Invalid location";

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ROUTING

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ROUTING

- Routing is bi-directional
  - Determines which Controller/Action or PageHandler to execute
  - Generates URLs within the application
- HTTP Verb is part of the route (MVC, WebAPI)
- In versioned API Services, the version is part of the route

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API SERVICES ROUTING

- Routes are independent of file structure or names
- The routing mechanism is attribute routing
  - Controllers - Route Attribute that combines tokens and literal values
  - Actions - Route Attribute or augment controller route with HTTP Verb attributes
  - Route attributes that begin with “/” reset the route
- Services support all Http Verbs

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HTTP FILES IN VS 2022

- The Visual Studio 2022 .http file editor provides a convenient way to test ASP.NET Core projects, especially API apps.
- The editor provides a UI that:
  - Creates and updates .http files.
  - Sends HTTP requests specified in .http files.
  - Displays the responses.
- https://learn.microsoft.com/en-us/aspnet/core/test/http-files?view=aspnetcore-8.0

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FILTERS

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FILTERS

- *Filters* in ASP.NET MVC allow you to run code before or after a particular stage in the execution pipeline.
  - Filters can be configured globally, per-controller, or per-action.
- Many types available
  - Action
  - Exception
  - Authorization
  - Resource
  - Result

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AVAILABLE FILTERS

- Authorization filters
  - Run first
  - Short circuit if not authorized
- Resource filters
  - Run before/after the rest of the pipeline
- Action filters
  - Run before/after action methods
- Exception filters
  - Run when an unhandled exception is thrown
- Result filters
  - Run before/after action results
- Can be set at the app, controller, or action level

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EXCEPTION FILTERS

- Come into play on unhandled exceptions

```
public class CustomExceptionFilterAttribute(
    IWebHostEnvironment hostEnvironment) : ExceptionFilterAttribute
{
  public override void OnException(ExceptionContext context)
  {
    string message = context.Exception.Message;
    context.Result = new BadRequestObjectResult(new {Message = message });
    //context.ExceptionHandled = true; //If this true, the exception is swallowed
  }
}
```

- Configured with AddControllers[WithViews] (for whole application filters)

```
builder.Services.AddControllers(config =>
{
   config.Filters.Add(new CustomExceptionFilterAttribute(builder.Environment));
})
```

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ACTION FILTERS

- Come into play before and/or after actions are executed

```
public class SimpleAuthenticationActionFilter : IActionFilter
{
        public void OnActionExecuting(ActionExecutingContext context)
        {   // do something before the action executes
        }
        public void OnActionExecuted(ActionExecutedContext context)
        {  // do something after the action executes
        }
}
```

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Questions?

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